AI Integrated Landslide Warning Mesh

Project Proposal

Team Name	Volt-Age
Category	University category
Theme	Public safety, innovation

Problem Definition

Introduction

Landslides are one of the major natural disasters frequently experienced in Sri Lanka. As we remember not so long ago we faced one of the most disastrous In the history of the country. Even though the government warn people about the occurrence of such disasters the general public do not obey them and we think we found the reason for that.

At present, there is no widely implemented and reliable system to warn people before a landslide occurs. This project is motivated by this gap and aims to identify potential landslide events in advance and provide early warnings to people in the affected areas to reduce the risk to human life.

Problem Analysis

The most effective way to minimize damage caused by landslides is to identify potential events before they occur. IoT technology enables continuous monitoring of critical parameters such as soil moisture content and ground tilt angle, while AI techniques can be used to detect unusual patterns and abnormal behavior in soil conditions. This approach aligns with Industry 5.0 principles by supporting human decision-making, reducing false alarms, and enhancing overall disaster preparedness.

Proposed Solution

Proposed Product

We propose a mesh with multiple nodes communicating with each other and identifying a land-slide before it even happens. Usually there are signs that nature shows before such a disaster happens one of them being the change of moisture in the soil and another one being the movement of middle layer of the soil.

We propose a node which has a soil moisture sensor and motion sensor which detects both of early warnings and send those data to a central hub,

which then being sent to a cloud server to analyze using weather reports' data in the area and artificial intelligence and determine the possibility of the land-slide.

We are planning to make this a commercial product by making a user friendly mobile application as well as a webserver so anyone with internet connectivity can use the feature.

Uniqueness of the Solution

Usually in Sri Lanka landslide warnings are predications; which is one of the main reasons general public ignore them because generally predictions are wrong. But with our proposed project it will gather data every time interval and analyze them to give more than 90% accurate warning.

And with enough upgrades done in the future with the help of the government may be we can implement in all the rural areas as well to warn the general public to move out of the way before unnecessary deaths occur.

Technical Overview and Implementation

Technical Details

The Node: *It will basically contain a soil moisture sensor, motion sensor (MPU6050), an esp32 s3 processor and a power source. For the power source we are thinking of using a li-ion battery as with limited usage of it, could last for Years with one charge as solar cells are not practical since sunlight might not guaranteed to every node.*
The main importance here is the mesh; as when one node collects its data from the sensors it creates a packet, label it and transmits them to the next node close to the central hub and itself (which is possible because nodes are strategically planted) and that node does the same and transmits both its packet and the one it received and so on until all nodes transfers their data to the central hub.

The Central Hub: *This is responsible for sending all the nodes' data to a cloud server. We are currently planning to use Wi-Fi for this but in the future it will be possible with a GSM module via mobile connectivity. Central Hub at the moment consists of a power source and a esp32 chip but with enough upgrades we can*

move onto raspberry-pi and do our AI calculation on the spot and send the refined data to the server.

The Server: *Currently we are doing our simulations on a free database (firebase) which can store data with each packets name.*

User Scenario

Imagine in a high-risk of landslide area an emergency notification of imminent land-slide will be sent to local phones about an hour before the actual disaster. The residencies will quickly move out of the area to safer locations.

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Additional Information

We are planning to use AI to make more accurate decisions using data in each node. For an instant there might be 2 nodes which the moisture sensors measure 2 types of soil types. So one reading might be normal for that soil type and with the help of AI we can determine if it is really concerning or not.

Simulations

We have started simulating the process with random data and 2 terminals to see if the codebase works properly and following are the results which emulates the nodes we are hoping to build

```
PS C:\Users\USER\OneDrive\Desktop\LDB> python node_b.py
>>
[Node B] Started
[Node B] Sent: {'node_id': 'B', 'soil': 78.12, 'tilt': 1.59, 'vibration': 0.033, 'battery': 3.69}
[Node B] Sent: {'node_id': 'B', 'soil': 78.89, 'tilt': 1.8, 'vibration': 0.02, 'battery': 4.08}
```

- This is data sending from nod b, which represents the farthest note from the central hub

```
python node_a.py
>> C:\Users\USER\OneDrive\Desktop\LDB>
[Node A] Started and listening for Node B
[Node A] Received from Node B: {'node_id': 'B', 'soil': 78.12, 'tilt': 1.59, 'vibration': 0.033, 'battery': 3.69}
[Node A] Forwarded combined data to Gateway

[Node A] Received from Node B: {'node_id': 'B', 'soil': 78.89, 'tilt': 1.8, 'vibration': 0.02, 'battery': 4.08}
[Node A] Forwarded combined data to Gateway

[Node A] Received from Node B: {'node_id': 'B', 'soil': 80.69, 'tilt': 1.27, 'vibration': 0.028, 'battery': 3.63}
[Node A] Forwarded combined data to Gateway

[Node A] Received from Node B: {'node_id': 'B', 'soil': 65.65, 'tilt': 1.18, 'vibration': 0.046, 'battery': 3.99}
[Node A] Forwarded combined data to Gateway
```

- This is the nod a terminal which represents the node closest to central hub. As seen here it receives data form node_b and generates its own data and combines them and send to the central hub(/gateway).

```
PS C:\Users\USER\OneDrive\Desktop\LDB> python gateway.py
>>
[Gateway] Listening...

========== GATEWAY RECEIVED ==========
Time: Wed Feb  4 15:35:02 2026
Node A: {'node_id': 'A', 'soil': 70.64, 'tilt': 0.74, 'vibration': 0.026, 'battery': 4.11}
Node B: {'node_id': 'B', 'soil': 78.12, 'tilt': 1.59, 'vibration': 0.033, 'battery': 3.69}
====================================

========== GATEWAY RECEIVED ==========
Time: Wed Feb  4 15:35:07 2026
Node A: {'node_id': 'A', 'soil': 72.42, 'tilt': 1.0, 'vibration': 0.013, 'battery': 3.95}
Node B: {'node_id': 'B', 'soil': 78.89, 'tilt': 1.8, 'vibration': 0.02, 'battery': 4.08}
====================================

========== GATEWAY RECEIVED ==========
Time: Wed Feb  4 15:35:12 2026
Node A: {'node_id': 'A', 'soil': 77.63, 'tilt': 0.64, 'vibration': 0.02, 'battery': 4.19}
Node B: {'node_id': 'B', 'soil': 80.69, 'tilt': 1.27, 'vibration': 0.028, 'battery': 3.63}
====================================
```

- This is the gateway terminal and all the data from node_b and node_a are visible.

Git hub repo for codebase and node enclosure designs: https://github.com/himashafdo/geo-mesh.git *(master branch)*